

Physics 513 QFT: HW 10

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I solve problem 2 before problem 1.

1. 2

I follow along with Sec. 13.3 in Schwartz. My notation comes from Schwartz and my QED notes. I introduce the muon field Ψ . It is the same as the electron field ψ except that the muon mass M is bigger than the electron mass m .

1.1. Lagrangian

The QED Lagrangian is

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1, \quad (1)$$

where the free part is

$$\mathcal{L}_0 = \bar{\psi}(i\not{\partial} - m)\psi + \bar{\Psi}(i\not{\partial} - M)\Psi - \frac{F^2}{4}. \quad (2)$$

The interaction Lagrangian is

$$\mathcal{L}_1 = -e\bar{\psi}\gamma^\mu\psi A_\mu - e\bar{\Psi}\gamma^\mu\Psi A_\mu. \quad (3)$$

Muons are just more massive electrons, so both fields couple to the electromagnetic field in the same way (since the coupling only depends on charge not mass). I think there might also be terms that mix muons and electrons, $\bar{\Psi}\psi$, but I don't have to worry about anything like that in this problem, so I ignore it.

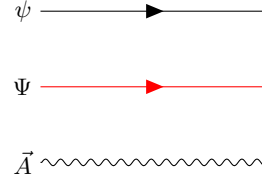


FIG. 1. Lines for different fields.

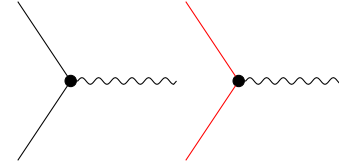


FIG. 2. Electron (left) and muon (right) QED vertices. Both vertices contribute the same factor: $-ie\gamma^\mu$.

1.2. Feynman Rules

I only give the Feynman rules I need for this problem. For example I don't need either the ψ or Ψ propagators so I ignore them. I use the symbols in Fig. 1 to represent the three fields. The photon propagator is

$$i\Pi_{\mu\nu}(\vec{p}) = -\frac{i\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon}, \quad (4)$$

where $\eta_{\mu\nu}$ is the Minkowski metric, $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$. Like Schwartz I am using the Feynman gauge. I show the two interaction vertices in Fig. 2.

1.3. $\mathcal{M}$

I assign the momenta

$$\vec{p}_1(e^-) + \vec{p}_2(e^+) = \vec{p}_3(\mu^-) + \vec{p}_4(\mu^+). \quad (5)$$

The total momenta are

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4. \quad (6)$$

Since this is a 2 to 2 reaction I use Mandelstam variables

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (7)$$

To leading order there is only one diagram (Fig. 3)

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \mathfrak{F}. \quad (8)$$

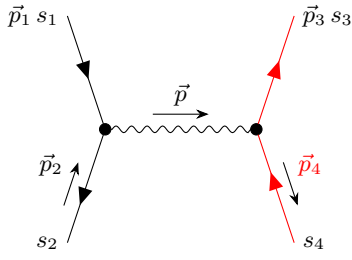


FIG. 3. Feynman diagram 3.

I use the shorthands

$$u_{s_n}(\vec{p}_n) = u_n, \quad v_{s_n}(\vec{p}_n) = v_n. \quad (9)$$

$$\mathfrak{F} = ie^2 \int d^4p \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon} \bar{u}_3 \gamma^\nu v_4 \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4) \quad (10)$$

$$\mathfrak{F} = ie^2 \int d^4p \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon} \bar{u}_3 \gamma^\nu v_4 \delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K}) \quad (11)$$

$$\mathfrak{F} = i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{\vec{P}^2} \bar{u}_3 \gamma^\nu v_4 \quad (12)$$

$$\mathfrak{F} = i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{s} \bar{u}_3 \gamma^\nu v_4 \quad (13)$$

$$i\delta^{(4)}(\vec{P} - \vec{K}) \mathcal{M} = i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{s} \bar{u}_3 \gamma^\nu v_4 \quad (14)$$

$$\mathcal{M} = e^2 \bar{v}_2 \gamma^\mu u_1 \frac{\eta_{\mu\nu}}{s} \bar{u}_3 \gamma^\nu v_4 \quad (15)$$

$$\mathcal{M} = \frac{e^2}{s} \bar{v}_2 \gamma^\mu u_1 \bar{u}_3 \eta_{\mu\nu} \gamma^\nu v_4 \quad (16)$$

$$\boxed{\mathcal{M} = \frac{e^2}{s} \bar{v}_2 \gamma^\mu u_1 \bar{u}_3 \gamma_\mu v_4} \quad (17)$$

1.4. $\mathcal{M}^\dagger$

$$\mathcal{M}^\dagger = \frac{e^2}{s} (\bar{v}_2 \gamma^\mu u_1 \bar{u}_3 \gamma_\mu v_4)^\dagger \quad (18)$$

$$\mathcal{M}^\dagger = \frac{e^2}{s} (\bar{u}_3 \gamma_\mu v_4)^\dagger (\bar{v}_2 \gamma^\mu u_1)^\dagger \quad (19)$$

While each term in parentheses is a scalar, the factors are matrices and spinors so switching the order is necessary.

$$\bar{v}_2 = v_2^\dagger \gamma^0 \quad (20)$$

$$(\bar{v}_2 \gamma^\mu u_1)^\dagger = (v_2^\dagger \gamma^0 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma^{\mu\dagger} \gamma^{0\dagger} v_2 \quad (21)$$

$$\gamma^{0\dagger} = \gamma^0, \quad \gamma^{a\dagger} = -\gamma^a \quad (22)$$

$$(\bar{v}_2 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma^{\mu\dagger} \gamma^0 v_2 \quad (23)$$

$$\gamma^{0\dagger} \gamma^0 = \gamma^0 \gamma^0 \quad (24)$$

$$\gamma^{a\dagger} \gamma^0 = -\gamma^a \gamma^0 \quad (25)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (26)$$

$$\{\gamma^0, \gamma^a\} = 2\eta^{0a} = 0 \implies \gamma^0 \gamma^a = -\gamma^a \gamma^0 \quad (27)$$

$$\gamma^{a\dagger} \gamma^0 = -\gamma^a \gamma^0 = \gamma^0 \gamma^a \quad (28)$$

$$\gamma^{0\dagger} \gamma^0 = \gamma^0 \gamma^0 \ \& \ \gamma^{a\dagger} \gamma^0 = \gamma^0 \gamma^a \implies \gamma^{\mu\dagger} \gamma^0 = \gamma^0 \gamma^\mu \quad (29)$$

$$(\bar{v}_2 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma^{\mu\dagger} \gamma^0 v_2 = u_1^\dagger \gamma^0 \gamma^\mu v_2 \quad (30)$$

$$\bar{u}_1 = u_1^\dagger \gamma^0 \quad (31)$$

$$(\bar{v}_2 \gamma^\mu u_1)^\dagger = \bar{u}_1 \gamma^\mu v_2 \quad (32)$$

In general for any two Dirac spinors ψ_1 and ψ_2

$$(\bar{\psi}_2 \gamma^\mu \psi_1)^\dagger = \bar{\psi}_1 \gamma^\mu \psi_2. \quad (33)$$

$$\mathcal{M}^\dagger = \frac{e^2}{s} (\bar{u}_3 \gamma_\mu v_4)^\dagger \bar{u}_1 \gamma^\mu v_2 \quad (34)$$

$$\gamma^\mu = (\gamma^0, \gamma^1, \gamma^2, \gamma^3) \quad (35)$$

$$\gamma_\mu = \eta_{\mu\nu} \gamma^\nu = (\gamma^0, -\gamma^1, -\gamma^2, -\gamma^3) \quad (36)$$

$$(\bar{\psi}_2 \gamma_0 \psi_1)^\dagger = (\bar{\psi}_2 \gamma^0 \psi_1)^\dagger = \bar{\psi}_1 \gamma_0 \psi_2 \quad (37)$$

$$(\bar{\psi}_2 \gamma_a \psi_1)^\dagger = -(\bar{\psi}_2 \gamma^a \psi_1)^\dagger = -\bar{\psi}_1 \gamma^a \psi_2 = \bar{\psi}_1 \gamma_a \psi_2 \quad (38)$$

So for two Dirac spinors I also have

$$(\bar{\psi}_2 \gamma_\mu \psi_1)^\dagger = \bar{\psi}_1 \gamma_\mu \psi_2. \quad (39)$$

$$\boxed{\mathcal{M}^\dagger = \frac{e^2}{s} \bar{v}_4 \gamma_\mu u_3 \bar{u}_1 \gamma^\mu v_2} \quad (40)$$

1.5. $|\mathcal{M}|^2$

$$|\mathcal{M}|^2 = \mathcal{M}\mathcal{M}^\dagger \quad (41)$$

$$|\mathcal{M}|^2 = \frac{e^4}{s^2} (\bar{v}_2 \gamma^\mu u_1) (\bar{u}_3 \gamma_\mu v_4) (\bar{v}_4 \gamma_\nu u_3) (\bar{u}_1 \gamma^\nu v_2) \quad (42)$$

1.6. Spin

I am calculating the unpolarized cross section meaning that no spin is preferred. The right way to do this is to sum over the final muon/ anti-muon spins and average over the initial electron/ positron spins.

$$|\mathcal{M}|^2 = \frac{e^4}{s^2} [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu u_{s_1}(\vec{p}_1)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu v_{s_4}(\vec{p}_4)] [\bar{v}_{s_4}(\vec{p}_4) \gamma_\nu u_{s_3}(\vec{p}_3)] [\bar{u}_{s_1}(\vec{p}_1) \gamma^\nu v_{s_2}(\vec{p}_2)] \quad (43)$$

$$\mathcal{M} = \frac{1}{4} \sum_{s_1 s_2 s_3 s_4=1}^2 |\mathcal{M}|^2 \quad (44)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_1 s_2 s_3 s_4=1}^2 [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu u_{s_1}(\vec{p}_1)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu v_{s_4}(\vec{p}_4)] [\bar{v}_{s_4}(\vec{p}_4) \gamma_\nu u_{s_3}(\vec{p}_3)] [\bar{u}_{s_1}(\vec{p}_1) \gamma^\nu v_{s_2}(\vec{p}_2)] \quad (45)$$

Scalars commute with everything so I can move them around as I wish.

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_1 s_2 s_3 s_4=1}^2 [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu u_{s_1}(\vec{p}_1)] [\bar{u}_{s_1}(\vec{p}_1) \gamma^\nu v_{s_2}(\vec{p}_2)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu v_{s_4}(\vec{p}_4) \bar{v}_{s_4}(\vec{p}_4) \gamma_\nu u_{s_3}(\vec{p}_3)] \quad (46)$$

I use the outer product

$$\sum_{s=1}^2 v_s(\vec{p}) \bar{v}_s(\vec{p}) = \not{p} - mI, \quad (47)$$

where I is the 4×4 identity matrix.

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_1 s_2 s_3=1}^2 [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu u_{s_1}(\vec{p}_1)] [\bar{u}_{s_1}(\vec{p}_1) \gamma^\nu v_{s_2}(\vec{p}_2)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu (\not{p}_4 - MI) \gamma_\nu u_{s_3}(\vec{p}_3)] \quad (48)$$

I use the other outer product identity

$$\sum_{s=1}^2 u_s(\vec{p}) \bar{u}_s(\vec{p}) = \not{p} + mI. \quad (49)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_1 s_2 s_3=1}^2 [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu u_{s_1}(\vec{p}_1) \bar{u}_{s_1}(\vec{p}_1) \gamma^\nu v_{s_2}(\vec{p}_2)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu (\not{p}_4 - MI) \gamma_\nu u_{s_3}(\vec{p}_3)] \quad (50)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 [\bar{v}_{s_2}(\vec{p}_2) \gamma^\mu (\not{p}_1 + mI) \gamma^\nu v_{s_2}(\vec{p}_2)] [\bar{u}_{s_3}(\vec{p}_3) \gamma_\mu (\not{p}_4 - MI) \gamma_\nu u_{s_3}(\vec{p}_3)] \quad (51)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 [\bar{v}_2 \gamma^\mu (\not{p}_1 + mI) \gamma^\nu v_2] [\bar{u}_3 \gamma_\mu (\not{p}_4 - MI) \gamma_\nu u_3] \quad (52)$$

I make I implicit.

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 [\bar{v}_2 \gamma^\mu (\not{p}_1 + m) \gamma^\nu v_2] [\bar{u}_3 \gamma_\mu (\not{p}_4 - M) \gamma_\nu u_3] \quad (53)$$

$$\mathcal{A} = \bar{v}_2 \gamma^\mu (\not{p}_1 + m) \gamma^\nu v_2, \quad \mathcal{B} = \bar{u}_3 \gamma_\mu (\not{p}_4 - M) \gamma_\nu u_3 \quad (54)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 \mathcal{A} \mathcal{B} \quad (55)$$

The spinors u and v are 4 component vectors, and the matrices $\gamma(\not{p} \pm m)\gamma$ are 4×4 . The products $\mathcal{A}$ and $\mathcal{B}$ are scalars. I introduce spinor component indices (like in my QED notes)

$$\mathcal{A} = (\bar{v}_2)_\alpha [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]_\beta^\alpha (v_2)^\beta, \quad (56)$$

where the spinor indices run over the 1,2,3,4 and I am using the Einstein summation convention. Because these are components I don't have to worry about commutation anymore and I can move factors as I wish.

$$\mathcal{A} = (v_2)^\beta (\bar{v}_2)_\alpha [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]_\beta^\alpha \quad (57)$$

Now comes a trick:

$$\mathcal{A} = (v_2 \bar{v}_2)_\alpha^\beta [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]_\beta^\alpha, \quad (58)$$

where $v_2 \bar{v}_2$ is a 4×4 matrix, which is why it has two components. $\mathcal{A}$ is still the same scalar (I checked in Mathematica) and everything is kosher.

$$\mathcal{B} = (\bar{u}_3)_\alpha [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]_\beta^\alpha (u_3)^\beta \quad (59)$$

$$\mathcal{B} = (u_3)^\beta (\bar{u}_3)_\alpha [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\alpha_\beta \quad (60)$$

$$\mathcal{B} = (u_3 \bar{u}_3)_\alpha^\beta [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\alpha_\beta \quad (61)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 (v_2 \bar{v}_2)_\alpha^\beta [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]^\alpha_\beta (u_3 \bar{u}_3)_\lambda^\rho [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\lambda_\rho \quad (62)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \sum_{s_2 s_3=1}^2 [v_{s_2}(\vec{p}_2) \bar{v}_{s_2}(\vec{p}_2)]_\alpha^\beta [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]^\alpha_\beta [u_{s_3}(\vec{p}_3) \bar{u}_{s_3}(\vec{p}_3)]_\lambda^\rho [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\lambda_\rho \quad (63)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[\sum_{s_2=1}^2 v_{s_2}(\vec{p}_2) \bar{v}_{s_2}(\vec{p}_2) \right]_\alpha^\beta [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]^\alpha_\beta \left[\sum_{s_3=1}^2 u_{s_3}(\vec{p}_3) \bar{u}_{s_3}(\vec{p}_3) \right]_\lambda^\rho [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\lambda_\rho \quad (64)$$

$$\mathcal{M} = \frac{e^4}{4s^2} (\not{p}_2 - m)_\alpha^\beta [\gamma^\mu (\not{p}_1 + m) \gamma^\nu]^\alpha_\beta (\not{p}_3 + M)_\lambda^\rho [\gamma_\mu (\not{p}_4 - M) \gamma_\nu]^\lambda_\rho \quad (65)$$

For any square matrices K & Q : $K_\alpha^\beta Q_\beta^\alpha = \text{Tr}(KQ)$.

$$\mathcal{M} = \frac{e^4}{4s^2} \text{Tr}[(\not{p}_2 - m) \gamma^\mu (\not{p}_1 + m) \gamma^\nu] \text{Tr}[(\not{p}_3 + M) \gamma_\mu (\not{p}_4 - M) \gamma_\nu] \quad (66)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \text{Tr}(\not{p}_2 \gamma^\mu \not{p}_1 \gamma^\nu - m \gamma^\mu \not{p}_1 \gamma^\nu + m \not{p}_2 \gamma^\mu \gamma^\nu - m^2 \gamma^\mu \gamma^\nu) \text{Tr}(\not{p}_3 \gamma_\mu \not{p}_4 \gamma_\nu - M \not{p}_3 \gamma_\mu \gamma_\nu + M \gamma_\mu \not{p}_4 \gamma_\nu - M^2 \gamma_\mu \gamma_\nu) \quad (67)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[\text{Tr}(\not{p}_2 \gamma^\mu \not{p}_1 \gamma^\nu) - m \text{Tr}(\gamma^\mu \not{p}_1 \gamma^\nu) + m \text{Tr}(\not{p}_2 \gamma^\mu \gamma^\nu) - m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \right] \left[\text{Tr}(\not{p}_3 \gamma_\mu \not{p}_4 \gamma_\nu) - M \text{Tr}(\not{p}_3 \gamma_\mu \gamma_\nu) + M \text{Tr}(\gamma_\mu \not{p}_4 \gamma_\nu) - M^2 \text{Tr}(\gamma_\mu \gamma_\nu) \right] \quad (68)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[\text{Tr}(\gamma_\alpha p_2^\alpha \gamma^\mu \gamma_\beta p_1^\beta \gamma^\nu) - m \text{Tr}(\gamma^\mu \gamma_\alpha p_1^\alpha \gamma^\nu) + m \text{Tr}(\gamma_\alpha p_2^\alpha \gamma^\mu \gamma^\nu) - m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \right] \left[\text{Tr}(\gamma_\alpha p_3^\alpha \gamma_\mu \gamma_\beta p_4^\beta \gamma_\nu) - M \text{Tr}(\gamma_\alpha p_3^\alpha \gamma_\mu \gamma_\nu) + M \text{Tr}(\gamma_\mu \gamma_\alpha p_4^\alpha \gamma_\nu) - M^2 \text{Tr}(\gamma_\mu \gamma_\nu) \right] \quad (69)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[p_2^\alpha p_1^\beta \text{Tr}(\gamma_\alpha \gamma^\mu \gamma_\beta \gamma^\nu) - m p_1^\alpha \text{Tr}(\gamma^\mu \gamma_\alpha \gamma^\nu) + m p_2^\alpha \text{Tr}(\gamma_\alpha \gamma^\mu \gamma^\nu) - m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \right] \left[p_3^\alpha p_4^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - M p_3^\alpha \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\nu) + M p_4^\alpha \text{Tr}(\gamma_\mu \gamma_\alpha \gamma_\nu) - M^2 \text{Tr}(\gamma_\mu \gamma_\nu) \right] \quad (70)$$

Schwartz Eq. (A.40) says

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu} \quad (71)$$

$$\text{Tr}(\gamma_\mu \gamma_\nu) = \text{Tr}(\eta_{\mu\alpha} \gamma^\alpha \eta_{\nu\beta} \gamma^\beta) \quad (72)$$

In this expression each γ^λ is a matrix but each $\eta_{\rho\sigma}$ is a single component (a scalar), so

$$\begin{aligned} \text{Tr}(\gamma_\mu \gamma_\nu) &= \eta_{\mu\alpha} \eta_{\nu\beta} \text{Tr}(\gamma^\alpha \gamma^\beta) = 4\eta_{\mu\alpha} \eta_{\nu\beta} \eta^{\alpha\beta} = 4\eta_{\mu\alpha} \eta_\nu^\alpha \\ &= 4\eta_{\mu\alpha} \delta_\nu^\alpha = 4\eta_{\mu\nu}. \end{aligned} \quad (73)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[p_2^\alpha p_1^\beta \text{Tr}(\gamma_\alpha \gamma^\mu \gamma_\beta \gamma^\nu) - m p_1^\alpha \text{Tr}(\gamma^\mu \gamma_\alpha \gamma^\nu) + m p_2^\alpha \text{Tr}(\gamma_\alpha \gamma^\mu \gamma^\nu) - 4m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - M p_3^\alpha \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\nu) + M p_4^\alpha \text{Tr}(\gamma_\mu \gamma_\alpha \gamma_\nu) - 4M^2 \eta_{\mu\nu} \right] \quad (74)$$

$$\mathcal{M} = \frac{e^4}{4s^2} \left[p_2^\alpha p_1^\beta \text{Tr}(\gamma_\alpha \gamma^\mu \gamma_\beta \gamma^\nu) - \cancel{m p_1^\alpha \text{Tr}(\gamma^\mu \gamma_\alpha \gamma^\nu)} + \cancel{m p_2^\alpha \text{Tr}(\gamma_\alpha \gamma^\mu \gamma^\nu)} - 4m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - \cancel{M p_3^\alpha \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\nu)} + \cancel{M p_4^\alpha \text{Tr}(\gamma_\mu \gamma_\alpha \gamma_\nu)} - 4M^2 \eta_{\mu\nu} \right] \quad (75)$$

From Schwartz (Eq. A.39): the trace of an odd number of gamma matrices is zero, $\text{Tr}(\gamma^\alpha \gamma^\beta \gamma^\mu) = 0$.

$$\mathcal{M} = \frac{e^4}{4s^2} \left[p_2^\alpha p_1^\beta \text{Tr}(\gamma_\alpha \gamma^\mu \gamma_\beta \gamma^\nu) - 4m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - 4M^2 \eta_{\mu\nu} \right] \quad (76)$$

Schwartz (Eq. A.41) is

$$\text{Tr}(\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu) = 4(\eta^{\alpha\mu} \eta^{\beta\nu} - \eta^{\alpha\beta} \eta^{\mu\nu} + \eta^{\alpha\nu} \eta^{\mu\beta}). \quad (77)$$

This immediately implies

$$\text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) = 4(\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}). \quad (78)$$

$$\mathcal{M} = \frac{4e^4}{s^2} \left[p_{2\alpha} p_{1\beta} (\eta^{\alpha\mu} \eta^{\beta\nu} - \eta^{\alpha\beta} \eta^{\mu\nu} + \eta^{\alpha\nu} \eta^{\mu\beta}) - m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta (\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) - M^2 \eta_{\mu\nu} \right] \quad (79)$$

$$\mathcal{M} = \frac{4e^4}{s^2} \left[p_2^\alpha p_1^\beta (\eta_\alpha^\mu \eta_\beta^\nu - \eta_{\alpha\beta} \eta^{\mu\nu} + \eta_\alpha^\nu \eta_\beta^\mu) - m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta (\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) - M^2 \eta_{\mu\nu} \right] \quad (80)$$

$$\mathcal{M} = \frac{4e^4}{s^2} \left[p_2^\alpha p_1^\beta (\delta_\alpha^\mu \delta_\beta^\nu - \eta_{\alpha\beta} \eta^{\mu\nu} + \delta_\alpha^\nu \delta_\beta^\mu) - m^2 \eta^{\mu\nu} \right] \left[p_3^\alpha p_4^\beta (\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) - M^2 \eta_{\mu\nu} \right] \quad (81)$$

$$\mathcal{M} = \frac{4e^4}{s^2} (p_2^\mu p_1^\nu - \eta^{\mu\nu} \vec{p}_1 \cdot \vec{p}_2 + p_2^\nu p_1^\mu - m^2 \eta^{\mu\nu}) (p_{3\mu} p_{4\nu} - \eta_{\mu\nu} \vec{p}_3 \cdot \vec{p}_4 + p_{3\nu} p_{4\mu} - M^2 \eta_{\mu\nu}) \quad (82)$$

I use the shorthand $p_{ij} = \vec{p}_i \cdot \vec{p}_j$.

$$\mathcal{M} = \frac{4e^4}{s^2} (p_2^\mu p_1^\nu - \eta^{\mu\nu} p_{12} + p_2^\nu p_1^\mu - m^2 \eta^{\mu\nu}) (p_{3\mu} p_{4\nu} - \eta_{\mu\nu} p_{34} + p_{3\nu} p_{4\mu} - M^2 \eta_{\mu\nu}) \quad (83)$$

$$\mathcal{M} = \frac{8e^4}{s^2} (p_{14} p_{23} + p_{13} p_{24} + M^2 p_{12} + m^2 p_{34} + 2m^2 M^2) \quad (84)$$

$$\eta^{\mu\nu} \eta_{\mu\nu} = 4 \quad (85)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = 2m^2 + 2p_{12} \\ s &= (\vec{p}_3 + \vec{p}_4)^2 = 2M^2 + 2p_{34} \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = m^2 + M^2 - 2p_{13} \\ t &= (\vec{p}_2 - \vec{p}_4)^2 = m^2 + M^2 - 2p_{24} \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = m^2 + M^2 - 2p_{14} \\ u &= (\vec{p}_2 - \vec{p}_3)^2 = m^2 + M^2 - 2p_{23} \\ p_{12} &= M^2 - m^2 + p_{34}, \quad p_{13} = p_{24}, \quad p_{14} = p_{23} \end{aligned} \quad (86)$$

$$\mathcal{M} = \frac{8e^4}{s^2} [p_{14}^2 + p_{24}^2 + (M^2 + m^2)p_{34} + M^4 + m^2 M^2] \quad (87)$$

$$2p_{34} = s - 2M^2, \quad 2p_{24} = m^2 + M^2 - t, \quad 2p_{14} = m^2 + M^2 - u \quad (88)$$

$$\begin{aligned} s + t + u &= 4m^2 + 2M^2 + 2\vec{p}_1 \cdot (\vec{p}_2 - \vec{p}_3 - \vec{p}_4) = \\ &= 4m^2 + 2M^2 + 2\vec{p}_1 \cdot (\vec{p}_2 - \vec{P}) = 4m^2 + 2M^2 - 2\vec{p}_1 \cdot \vec{p}_1 \\ &= 2m^2 + 2M^2 \end{aligned} \quad (89)$$

$$\mathcal{M} = 2 \frac{e^4}{s^2} [t^2 + u^2 + 4s(m^2 + M^2) - 2(m^2 + M^2)^2] \quad (90)$$

1.7. Momenta

I work in the center of mass frame where the initial momenta are

$$\vec{p}_1 = (E, \mathbf{p}), \quad \vec{p}_2 = (E, -\mathbf{p}), \quad E = \sqrt{m^2 + p^2}, \quad p = |\mathbf{p}|. \quad (91)$$

The final momenta are

$$\vec{p}_3 = (\mathcal{E}, \mathbf{k}), \quad \vec{p}_4 = (\mathcal{E}, -\mathbf{k}), \quad \mathcal{E} = \sqrt{M^2 + k^2}, \quad k = |\mathbf{k}|. \quad (92)$$

conservation of energy implies

$$2E = 2\mathcal{E} \implies E = \mathcal{E}. \quad (93)$$

Without loss of generality I take

$$\mathbf{p} = p \hat{\mathbf{z}}, \quad \mathbf{k} = k(\sin \theta, 0, \cos \theta). \quad (94)$$

So the final and initial momenta are

$$\begin{aligned} \vec{p}_1 &= (E, 0, 0, p) & \vec{p}_2 &= (E, 0, 0, -p) \\ \vec{p}_3 &= (E, k \sin \theta, 0, k \cos \theta) \\ \vec{p}_4 &= (E, -k \sin \theta, 0, -k \cos \theta) \\ E &= \sqrt{M^2 + k^2} = \sqrt{m^2 + p^2}. \end{aligned} \quad (95)$$

With these definitions I calculate s , t , and u .

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = 2m^2 + 2\vec{p}_1 \cdot \vec{p}_2 = 2m^2 + 2\vec{p}_1 \cdot \vec{p}_2 \\ &= 2m^2 + 2(E^2 - \mathbf{p}_1 \cdot \mathbf{p}_2) = 2m^2 + 2E^2 + 2p^2 = \\ &= 2E^2 + 2(m^2 + p^2) = 2E^2 + 2E^2 = 4E^2 \end{aligned} \quad (96)$$

$$\boxed{s = 4E^2} \quad (97)$$

$$t = (\vec{p}_1 - \vec{p}_3)^2 = m^2 + M^2 - 2\vec{p}_1 \cdot \vec{p}_3 \quad (98)$$

$$\boxed{t = m^2 + M^2 - 2E^2 + 2pk \cos \theta} \quad (99)$$

$$u = (\vec{p}_1 - \vec{p}_4)^2 = m^2 + M^2 - 2\vec{p}_1 \cdot \vec{p}_4 \quad (100)$$

$$\boxed{u = m^2 + M^2 - 2E^2 - 2pk \cos \theta} \quad (101)$$

1.8. Differential Cross Section

The formula for the differential cross section in the center of mass (CM) frame is Schwartz Eq. (5.32)

$$\frac{d\sigma}{d\Omega} = \frac{\mathcal{M}}{64\pi^2 E_{\text{CM}}^2} \frac{p_f}{p_i}, \quad (102)$$

where $E_{\text{CM}} = 2E$ is the total energy in the CM frame, $p_i = |\mathbf{p}_1| = |\mathbf{p}_2| = p$, and $p_f = |\mathbf{p}_3| = |\mathbf{p}_4| = k$.

$$\frac{d\sigma}{d\Omega} = \frac{\mathcal{M}}{16(4\pi)^2 E^2} \frac{k}{p} \quad (103)$$

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{8(4\pi)^2 E^2 s^2} \frac{k}{p} [t^2 + u^2 + 4s(m^2 + M^2) - 2(m^2 + M^2)^2] \quad (104)$$

The fine structure constant is

$$\alpha = \frac{e^2}{4\pi}. \quad (105)$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{8E^2 s^2} \frac{k}{p} [t^2 + u^2 + 4s(m^2 + M^2) - 2(m^2 + M^2)^2] \quad (106)$$

$$s = 4E^2 \quad (107)$$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{16E^6 8} \frac{k}{p} [t^2 + u^2 + 16E^2(m^2 + M^2) - 2(m^2 + M^2)^2] \quad (108)$$

$$\begin{aligned} t &= m^2 + M^2 - 2E^2 + 2pk \cos \theta \\ u &= m^2 + M^2 - 2E^2 - 2pk \cos \theta \end{aligned} \quad (109)$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{16E^6} \frac{k}{p} (E^4 + k^2 p^2 \cos^2 \theta + E^2(m^2 + M^2))} \quad (110)$$

This is identical to Eq. (13.75) in Schwartz (except we swap k and p).

2. 1

2.1. $\mathcal{M}$

I calculated $\mathcal{M}$ for this process in HW 9. I copy my results from there.

I relabel $\gamma' \rightarrow G$. The interaction Lagrangian is

$$\mathcal{L}_1 = g\bar{\psi}\psi\phi + iG\bar{\psi}\gamma_5\psi\pi. \quad (111)$$

$$\vec{p}_1(e^-) + \vec{k}(\phi) \rightarrow \vec{p}_2(e^-) + \vec{q}(\pi) \quad (112)$$

$$\vec{P} = \vec{p}_1 + \vec{k} = \vec{p}_2 + \vec{q} \quad (113)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{k})^2 = (\vec{p}_2 + \vec{q})^2, \\ t &= (\vec{p}_1 - \vec{p}_2)^2 = (\vec{k} - \vec{q})^2, \\ u &= (\vec{p}_1 - \vec{q})^2 = (\vec{k} - \vec{p}_2)^2. \end{aligned} \quad (114)$$

$$\mathcal{M} = gG\bar{u}_2 \left(\gamma_5 \frac{\not{p} + m}{s - m^2} + \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 \right) u_1 \quad (115)$$

The electron mass is m .

$$u_{s_n}(\vec{p}_n) = u_n \quad (116)$$

$$\frac{\mathcal{M}}{gG} = \bar{u}_2 \gamma_5 \frac{\not{p} + m}{s - m^2} u_1 + \bar{u}_2 \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 u_1 \quad (117)$$

$$\frac{\mathcal{M}}{gG} = \bar{u}_2 \gamma_5 \frac{\not{p}_2 + \not{q} + m}{s - m^2} u_1 + \bar{u}_2 \frac{\not{p}_1 - \not{q} + m}{u - m^2} \gamma_5 u_1 \quad (118)$$

$$\frac{\mathcal{M}}{gG} = \bar{u}_2 \gamma_5 \frac{\gamma_\mu p_2^\mu + \not{q} + m}{s - m^2} u_1 + \bar{u}_2 \frac{\gamma_\mu p_1^\mu - \not{q} + m}{u - m^2} \gamma_5 u_1 \quad (119)$$

$$\begin{aligned} \frac{\mathcal{M}}{gG} &= \bar{u}_2 \frac{\gamma_5 \gamma_\mu p_2^\mu + \gamma_5 \not{q} + \gamma_5 m}{s - m^2} u_1 \\ &\quad + \bar{u}_2 \frac{\gamma_\mu p_1^\mu \gamma_5 - \not{q} \gamma_5 + m \gamma_5}{u - m^2} u_1 \end{aligned} \quad (120)$$

$$\begin{aligned} \frac{\mathcal{M}}{gG} &= \bar{u}_2 \frac{\gamma_5 \gamma_\mu p_2^\mu + \gamma_5 \not{q} + \gamma_5 m}{s - m^2} u_1 \\ &\quad + \bar{u}_2 \frac{\gamma_\mu \gamma_5 p_1^\mu - \not{q} \gamma_5 + m \gamma_5}{u - m^2} u_1 \end{aligned} \quad (121)$$

$$\{\gamma_5, \gamma_\mu\} = 0 \implies \gamma_5 \gamma_\mu = -\gamma_\mu \gamma_5 \quad (122)$$

$$\frac{\mathcal{M}}{gG} = \bar{u}_2 \frac{-\gamma_\mu \gamma_5 p_2^\mu + \gamma_5 \not{p}_2 + \gamma_5 m}{s - m^2} u_1 + \bar{u}_2 \frac{-\gamma_5 \gamma_\mu p_1^\mu - \not{p}_1 \gamma_5 + m \gamma_5}{u - m^2} u_1 \quad (123)$$

$$\frac{\mathcal{M}}{gG} = \bar{u}_2 \frac{\gamma_5 \not{p}_2 - (\not{p}_2 - m) \gamma_5}{s - m^2} u_1 - \bar{u}_2 \frac{\gamma_5 (\not{p}_1 - m) + \not{p}_1 \gamma_5}{u - m^2} u_1 \quad (124)$$

In HW 8 I derived

$$(\not{p} - m) u_s(\vec{p}) = \bar{u}_s(\vec{p})(\not{p} - m) = 0. \quad (125)$$

$$\frac{\mathcal{M}}{gG} = \frac{\bar{u}_2 \gamma_5 \not{p}_2 - \bar{u}_2 (\not{p}_2 - m) \gamma_5}{s - m^2} u_1 - \bar{u}_2 \frac{\gamma_5 (\not{p}_1 - m) u_1 + \not{p}_1 \gamma_5 u_1}{u - m^2} \quad (126)$$

$$\frac{\mathcal{M}}{gG} = \frac{\bar{u}_2 \gamma_5 \not{p}_2}{s - m^2} u_1 - \bar{u}_2 \frac{\not{p}_1 \gamma_5 u_1}{u - m^2} \quad (127)$$

$$\{\gamma_5, \gamma_\mu\} = 0 \implies \gamma_5 \gamma_\mu = -\gamma_\mu \gamma_5 \implies \gamma_5 \not{p} = -\not{p} \gamma_5 \quad (128)$$

$$\frac{\mathcal{M}}{gG} = \frac{\bar{u}_2 \gamma_5 \not{p}_2}{s - m^2} u_1 + \bar{u}_2 \frac{\gamma_5 \not{p}_1 u_1}{u - m^2} \quad (129)$$

$$\frac{\mathcal{M}}{gG} = \frac{\bar{u}_2 \gamma_5 \not{p}_1 u_1}{s - m^2} + \frac{\bar{u}_2 \gamma_5 \not{p}_2 u_1}{u - m^2} \quad (130)$$

$$\mathcal{M} = gG \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) \bar{u}_2 \gamma_5 \not{p}_1 u_1 \quad (131)$$

2.2. $\mathcal{M}^\dagger$

$$\mathcal{M}^\dagger = gG \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) (\bar{u}_2 \gamma_5 \not{p}_1 u_1)^\dagger \quad (132)$$

$$\mathcal{M}^\dagger = gG \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) q_\mu (\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger \quad (133)$$

$$(\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger = (u_2^\dagger \gamma_0 \gamma_5 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma^{\mu\dagger} \gamma_5^\dagger \gamma_0^\dagger u_2 \quad (134)$$

$$\gamma_0^\dagger = \gamma_0, \quad \gamma_5^\dagger = \gamma_5 \quad (135)$$

$$(\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma^{\mu\dagger} \gamma_5 \gamma_0 u_2 \quad (136)$$

$$\gamma_0 \gamma_0 = I \quad (137)$$

$$(\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger = u_1^\dagger \gamma_0 \gamma_0 \gamma^{\mu\dagger} \gamma_5 \gamma_0 u_2 = -u_1^\dagger \gamma_0 \gamma_0 \gamma^{\mu\dagger} \gamma_0 \gamma_5 u_2 \quad (138)$$

From my QED notes

$$\gamma_0 \gamma^{\mu\dagger} \gamma_0 = \gamma^\mu \quad (139)$$

$$(\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger = -u_1^\dagger \gamma_0 \gamma^\mu \gamma_5 u_2 \quad (140)$$

$$\bar{u} = u^\dagger \gamma_0 \quad (141)$$

$$(\bar{u}_2 \gamma_5 \gamma^\mu u_1)^\dagger = -\bar{u}_1 \gamma^\mu \gamma_5 u_2 = \bar{u}_1 \gamma_5 \gamma^\mu u_2 \quad (142)$$

$$\mathcal{M}^\dagger = gG \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) q_\mu \bar{u}_1 \gamma_5 \gamma^\mu u_2 \quad (143)$$

$$\mathcal{M}^\dagger = gG \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) \bar{u}_1 \gamma_5 \not{p}_1 u_2 \quad (144)$$

2.3. $|\mathcal{M}|^2$

$$|\mathcal{M}|^2 = \mathcal{M} \mathcal{M}^\dagger \quad (145)$$

$$|\mathcal{M}|^2 = \left(\frac{gG}{s - m^2} + \frac{gG}{u - m^2} \right)^2 (\bar{u}_2 \gamma_5 \not{p}_1 u_1) (\bar{u}_1 \gamma_5 \not{p}_1 u_2) \quad (146)$$

2.4. Spin

I want the unpolarized cross section. So I average over the initial spin and sum over the final spin.

$$\mathcal{M} = \frac{1}{2} \sum_{s_1 s_2=1}^2 |\mathcal{M}|^2 \quad (147)$$

$$\mathcal{M} = \frac{g^2 G^2}{2} \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right)^2 \mathcal{B} \quad (148)$$

$$\mathcal{B} = \sum_{s_1 s_2=1}^2 (\bar{u}_2 \gamma_5 \not{p}_1 u_1) (\bar{u}_1 \gamma_5 \not{p}_1 u_2) \quad (149)$$

$$\mathcal{B} = \sum_{s_1 s_2=1}^2 \bar{u}_2 \gamma_5 \not{p}_1 u_1 \bar{u}_1 \gamma_5 \not{p}_1 u_2 \quad (150)$$

$$\mathcal{B} = \sum_{s_2=1}^2 \bar{u}_2 \gamma_5 \not{q} \left(\sum_{s_1=1}^2 u_1 \bar{u}_1 \right) \gamma_5 \not{q} u_2 \quad (151)$$

$$\mathcal{B} = \sum_{s_2=1}^2 \bar{u}_2 \gamma_5 \not{q} \left[\sum_{s_1=1}^2 u_{s_1}(\vec{p}_1) \bar{u}_{s_1}(\vec{p}_1) \right] \gamma_5 \not{q} u_2 \quad (152)$$

$$\sum_{s=1}^2 u_s(\vec{p}) \bar{u}_s(\vec{p}) = \not{p} + m. \quad (153)$$

$$\mathcal{B} = \sum_{s_2=1}^2 \bar{u}_2 \gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q} u_2 \quad (154)$$

I start using spinor component notation.

$$\mathcal{B} = \sum_{s_2=1}^2 (\bar{u}_2)_\alpha [\gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q}]_\beta^\alpha (u_2)^\beta \quad (155)$$

$$\mathcal{B} = \sum_{s_2=1}^2 (u_2)^\beta (\bar{u}_2)_\alpha [\gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q}]_\beta^\alpha \quad (156)$$

$$\mathcal{B} = \sum_{s_2=1}^2 (u_2 \bar{u}_2)_\alpha^\beta [\gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q}]_\beta^\alpha \quad (157)$$

$$\mathcal{B} = \left[\sum_{s_2=1}^2 u_{s_2}(\vec{p}_2) \bar{u}_{s_2}(\vec{p}_2) \right]_\alpha^\beta [\gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q}]_\beta^\alpha \quad (158)$$

$$\mathcal{B} = (\not{p}_2 + m)_\alpha^\beta [\gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q}]_\beta^\alpha \quad (159)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_2 + m) \gamma_5 \not{q} (\not{p}_1 + m) \gamma_5 \not{q} \right] \quad (160)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_2 + m) (\gamma_5 \not{p}_1 \gamma_5 \not{q} + m \gamma_5 \not{q}) \right] \quad (161)$$

$$\gamma_5 \not{p} = -\not{p} \gamma_5 \quad (162)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_2 + m) \gamma_5^2 (\not{p}_1 \not{q} - m \not{q}) \right] \quad (163)$$

From my QED notes

$$\gamma_5^2 = I. \quad (164)$$

$$\mathcal{B} = \text{Tr} \left[(\not{p}_2 + m) (\not{p}_1 \not{q} - m \not{q}^2) \right] \quad (165)$$

$$\mathcal{B} = \text{Tr} \left(\not{p}_2 \not{p}_1 \not{q} + m \not{p}_1 \not{q} - m \not{p}_2 \not{q}^2 - m^2 \not{q}^2 \right) \quad (166)$$

$$\mathcal{B} = \text{Tr}(\not{p}_2 \not{p}_1 \not{q}) + m \text{Tr}(\not{p}_1 \not{q}) - m \text{Tr}(\not{p}_2 \not{q}^2) - m^2 \text{Tr}(\not{q}^2) \quad (167)$$

$$\text{Tr}(\gamma^\alpha \gamma^\beta \gamma^\mu) = 0 \implies \text{Tr}(\not{p} \not{k} \not{q}) = 0 \quad (168)$$

$$\mathcal{B} = \text{Tr}(\not{p}_2 \not{p}_1 \not{q}) - m^2 \text{Tr}(\not{q}^2) \quad (169)$$

$$\mathcal{B} = p_2^\alpha q^\mu p_1^\beta q^\nu \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - m^2 q^\mu q^\nu \text{Tr}(\gamma_\mu \gamma_\nu) \quad (170)$$

$$\text{Tr}(\gamma_\mu \gamma_\nu) = 4 \eta_{\mu\nu} \quad (171)$$

$$\text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) = 4(\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) \quad (172)$$

$$\mathcal{B} = p_2^\alpha q^\mu p_1^\beta q^\nu \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - 4m^2 q^\mu q^\nu \eta_{\mu\nu} \quad (173)$$

I use the shorthand

$$P_{ij} = \vec{p}_i \cdot \vec{p}_j, \quad p_{iq} = \vec{p}_i \cdot \vec{q}. \quad (174)$$

$$\mathcal{B} = p_2^\alpha q^\mu p_1^\beta q^\nu \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) - 4m^2 p_{qq} \quad (175)$$

$$\frac{\mathcal{B}}{4} = p_2^\alpha q^\mu p_1^\beta q^\nu (\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) - m^2 p_{qq} \quad (176)$$

$$\frac{\mathcal{B}}{4} = p_{1q} p_{2q} - p_{12} p_{qq} + p_{1q} p_{2q} - m^2 p_{qq} \quad (177)$$

$$\frac{\mathcal{B}}{4} = 2p_{1q} p_{2q} - \cancel{p_{12} p_{qq}} - \cancel{m^2 p_{qq}} \quad (178)$$

The scalar fields are massless so

$$p_{qq} = \vec{q} \cdot \vec{q} = 0. \quad (179)$$

$$\mathcal{B} = 8p_{1q} p_{2q} \quad (180)$$

$$\mathcal{B} = 8(\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q}) \quad (181)$$

$$\mathcal{M} = 4g^2 G^2 \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right)^2 (\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q}) \quad (182)$$

$$s = (\vec{p}_2 + \vec{q})^2 = (\vec{p}_2)^2 + \cancel{(\vec{q})^2} + 2\vec{p}_2 \cdot \vec{q} = m^2 + 2\vec{p}_2 \cdot \vec{q} \quad (183)$$

$$u = (\vec{p}_1 - \vec{q})^2 = m^2 - 2\vec{p}_1 \cdot \vec{q} \quad (184)$$

$$\mathcal{M} = g^2 G^2 \left(\frac{1}{\vec{p}_2 \cdot \vec{q}} - \frac{1}{\vec{p}_1 \cdot \vec{q}} \right)^2 (\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q}) \quad (185)$$

$$\mathcal{M} = g^2 G^2 \left[\frac{(\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q})}{(\vec{p}_2 \cdot \vec{q})^2} + \frac{(\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q})}{(\vec{p}_1 \cdot \vec{q})^2} - 2 \frac{(\vec{p}_1 \cdot \vec{q})(\vec{p}_2 \cdot \vec{q})}{(\vec{p}_2 \cdot \vec{q})(\vec{p}_1 \cdot \vec{q})} \right] \quad (186)$$

$$\mathcal{M} = g^2 G^2 \left(\frac{\vec{p}_1 \cdot \vec{q}}{\vec{p}_2 \cdot \vec{q}} + \frac{\vec{p}_2 \cdot \vec{q}}{\vec{p}_1 \cdot \vec{q}} - 2 \right) \quad (187)$$

$$\begin{aligned} s &= (\vec{p}_1 + \vec{k})^2 = m^2 + 2\vec{p}_1 \cdot \vec{k} \\ s &= (\vec{p}_2 + \vec{q})^2 = m^2 + 2\vec{p}_2 \cdot \vec{q} \\ \vec{p}_1 \cdot \vec{k} &= \vec{p}_2 \cdot \vec{q} \end{aligned} \quad (188)$$

$$\mathcal{M} = g^2 G^2 \left(\frac{\vec{p}_1 \cdot \vec{q}}{\vec{p}_1 \cdot \vec{k}} + \frac{\vec{p}_2 \cdot \vec{q}}{\vec{p}_2 \cdot \vec{q}} - 2 \right) \quad (189)$$

2.5. Kinematics

I work in the frame with initial momenta

$$\vec{p}_1 = (m, 0, 0, 0), \quad \vec{k} = (\omega, 0, 0, \omega) \quad (190)$$

in Cartesian coordinates. The final momenta are

$$\vec{q} = (\omega', \mathbf{q}), \quad \vec{p}_2 = (E, \mathbf{p}_2). \quad (191)$$

$$\omega' = q = |\mathbf{q}| \quad (192)$$

$$E = \sqrt{m^2 + p_2^2}, \quad p_2 = |\mathbf{p}_2| \quad (193)$$

Without loss of generality I can orient my coordinates so that

$$\vec{q} = (\omega', \omega' \sin \theta, 0, \omega' \cos \theta) \quad (194)$$

$$\vec{p}_2 = (E, p_2^x, 0, p_2^z). \quad (195)$$

I use conservation of 4-momentum.

$$p_2^x + \omega' \sin \theta = 0 \implies p_2^x = -\omega' \sin \theta \quad (196)$$

$$\omega = \omega' \cos \theta + p_2^z \implies p_2^z = \omega - \omega' \cos \theta \quad (197)$$

$$\vec{p}_2 = (E, -\omega' \sin \theta, 0, \omega - \omega' \cos \theta) \quad (198)$$

$$\begin{aligned} p_2^2 &= (p_2^x)^2 + (p_2^z)^2 = \omega'^2 \sin^2 \theta + (\omega - \omega' \cos \theta)^2 = \\ &= \omega'^2 \sin^2 \theta + \omega^2 + \omega'^2 \cos^2 \theta - 2\omega\omega' \cos \theta = \\ &= \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta \end{aligned} \quad (199)$$

$$p_2 = \sqrt{\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta} \quad (200)$$

$$E = \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta} \quad (201)$$

$$m + \omega = E + \omega' \quad (202)$$

$$E = m + \omega - \omega' \quad (203)$$

$$E^2 = (m + \omega - \omega')^2 \quad (204)$$

This implies (I worked it out with Mathematica)

$$\omega' = \frac{m\omega}{m + \omega(1 - \cos \theta)}. \quad (205)$$

$$\mathcal{M} = g^2 G^2 \left(\frac{\vec{p}_1 \cdot \vec{q}}{\vec{p}_1 \cdot \vec{k}} + \frac{\vec{p}_2 \cdot \vec{q}}{\vec{p}_2 \cdot \vec{q}} - 2 \right) \quad (206)$$

$$\vec{p}_1 \cdot \vec{k} = m\omega, \quad \vec{p}_1 \cdot \vec{q} = m\omega' \quad (207)$$

$$\mathcal{M} = g^2 G^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 \right) \quad (208)$$

2.6. Cross Section

I get the cross section from Eqs. (5.21) and (5.22) in Schwartz.

$$d\sigma = \frac{\mathcal{M}}{64\pi^2 m \omega \omega' E} \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}_2 - \vec{q}) d^3 p_2 d^3 q \quad (209)$$

$$64\pi^2 m \omega d\sigma = \frac{\mathcal{M}}{\omega' E} \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}_2 - \vec{q}) d^3 p_2 d^3 q \quad (210)$$

$$64\pi^2 m \omega \sigma = \int d^3 p_2 \int d^3 q \frac{\mathcal{M}}{\omega' E} \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}_2 - \vec{q}) \quad (211)$$

$$\begin{aligned} \frac{64\pi^2 m \omega}{g^2 G^2} \sigma &= \int d^3 q \int \frac{d^3 p_2}{\omega' E} \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 \right) \\ &\quad \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}_2 - \vec{q}) \end{aligned} \quad (212)$$

$$\omega' = \omega'(q) = q, \quad E = E(p_2) = \sqrt{m^2 + p_2^2} \quad (213)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int d^3 q \int \frac{d^3 p_2}{q E(p_2)} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p}_2 - \vec{q}) \quad (214)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int d^3 q \int \frac{d^3 p}{q E(p)} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p} - \vec{q}) \quad (215)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int \frac{d^3 q}{q} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \int \frac{d^3 p}{E(p)} \delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p} - \vec{q}) \quad (216)$$

$$\delta^{(4)}(\vec{p}_1 + \vec{k} - \vec{p} - \vec{q}) = \delta^{(3)}(\mathbf{p}_1 + \mathbf{k} - \mathbf{p} - \mathbf{q}) \delta(m + \omega - E - \omega') \quad (217)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int \frac{d^3 q}{q} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \int \frac{d^3 p}{E(p)} \delta^{(3)}(\mathbf{p}_1 + \mathbf{k} - \mathbf{p} - \mathbf{q}) \delta[m + \omega - E(p) - \omega'] \quad (218)$$

$$\omega' = q, \quad \mathbf{p}_1 = \mathbf{0}, \quad \mathbf{k} = \omega \hat{\mathbf{z}} \quad (219)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int \frac{d^3 q}{q} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \int \frac{d^3 p}{E(p)} \delta^{(3)}(\omega \hat{\mathbf{z}} - \mathbf{p} - \mathbf{q}) \delta[m + \omega - E(p) - q] \quad (220)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int \frac{d^3 q}{q} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \int \frac{d^3 p}{E(p)} \delta^{(3)}(\omega \hat{\mathbf{z}} - \mathbf{p} - \mathbf{q}) \delta[m + \omega - E(p) - q] \quad (221)$$

$$\mathbf{0} = \omega \hat{\mathbf{z}} - \mathbf{p} - \mathbf{q} \implies \mathbf{p} = \omega \hat{\mathbf{z}} - \mathbf{q} \implies p^2 = \omega^2 + q^2 - 2\omega \hat{\mathbf{z}} \cdot \mathbf{q} = \omega^2 + q^2 - 2\omega q_z = \omega^2 + q^2 - 2\omega q \cos \theta \quad (222)$$

Remember θ is the angle from the z axis (along which the collision occurred). This matches up with θ as conventionally defined in spherical polar coordinates. Which is why I

set things up this way.

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int \frac{d^3 q}{q} \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \int \frac{d^3 p}{\sqrt{m^2 + p^2}} \delta^{(3)}(\omega \hat{\mathbf{z}} - \mathbf{p} - \mathbf{q}) \delta[m + \omega - \sqrt{m^2 + p^2} - q] \quad (223)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int d^3 q \left(\frac{q}{\omega} + \frac{\omega}{q} - 2 \right) \frac{\delta[m + \omega - \sqrt{m^2 + \omega^2 + q^2 - 2\omega q \cos \theta} - q]}{q \sqrt{m^2 + \omega^2 + q^2 - 2\omega q \cos \theta}} \quad (224)$$

I change the dummy variables to exploit the spherical symmetry in this integral.

$$\frac{64\pi^2 m\omega}{g^2 G^2} \sigma = \int_0^{4\pi} d\Omega \int_0^\infty dr r^2 \left(\frac{r}{\omega} + \frac{\omega}{r} - 2 \right) \frac{\delta[m + \omega - \sqrt{m^2 + \omega^2 + r^2 - 2\omega r \cos \theta} - r]}{r \sqrt{m^2 + \omega^2 + r^2 - 2\omega r \cos \theta}} \quad (225)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} d\sigma = d\Omega \int_0^\infty dr \left(\frac{r}{\omega} + \frac{\omega}{r} - 2 \right) \frac{r \delta[m + \omega - \sqrt{m^2 + \omega^2 + r^2 - 2\omega r \cos \theta} - r]}{\sqrt{m^2 + \omega^2 + r^2 - 2\omega r \cos \theta}} \quad (226)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \frac{d\sigma}{d\Omega} = \int_0^\infty dx \left(\frac{x}{\omega} + \frac{\omega}{x} - 2 \right) \frac{x \delta[m + \omega - \sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta} - x]}{\sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta}} \quad (227)$$

$$f(x) = m + \omega - \sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta} - x \quad (228)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \frac{d\sigma}{d\Omega} = \int_0^\infty dx \left(\frac{x}{\omega} + \frac{\omega}{x} - 2 \right) \frac{x \delta[f(x)]}{\sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta}} \quad (229)$$

$f(x)$ has only one zero in the range of integration:

$$f(x_0) = 0, \quad x_0 = \omega' = \frac{m\omega}{m + \omega(1 - \cos \theta)}, \quad (230)$$

where I used Eq. 205. I use the identity

$$\delta[f(x)] = \frac{\delta(x - \omega')}{|f'(\omega')|}. \quad (231)$$

$$|f'(\omega')| = \frac{m\omega}{\omega'(m + \omega - \omega')} \quad (232)$$

$$\frac{64\pi^2 m\omega}{g^2 G^2} \frac{d\sigma}{d\Omega} = \frac{1}{|f'(\omega')|} \int_0^\infty dx \left(\frac{x}{\omega} + \frac{\omega}{x} - 2 \right) \frac{x \delta(x - \omega')}{\sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta}} \quad (233)$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{gG}{8\pi m\omega} \right)^2 \omega'(m + \omega - \omega') \int_0^\infty dx \left(\frac{x}{\omega} + \frac{\omega}{x} - 2 \right) \frac{x \delta(x - \omega')}{\sqrt{m^2 + \omega^2 + x^2 - 2\omega x \cos \theta}} \quad (234)$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{gG}{8\pi m\omega} \right)^2 \omega'(m + \omega - \omega') \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 \right) \frac{\omega'}{\sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta}} \quad (235)$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{gG\omega'}{8\pi m\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 \right) \frac{m + \omega - \omega'}{\sqrt{(m + \omega - \omega')^2}} \quad (236)$$

Conservation of energy yields $m + \omega - \omega' > 0$.

$$\frac{d\sigma}{d\Omega} = \left(\frac{gG\omega'}{8\pi m\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 \right) \quad (237)$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\omega'}{\omega} \left[\frac{gG}{8\pi m} \left(1 - \frac{\omega'}{\omega} \right) \right]^2} \quad (238)$$